

The Three Questions

Immanuel Kant, *Critique of Pure Reason* (1781)

Carlos Cotrini

A land of truth in a sea of illusion



C. D. Friedrich,

Wanderer above the Sea of Fog (c. 1817)

Es ist das **Land der Wahrheit** (ein reizender Name), umgeben von einem weiten und stürmischen Oceane, dem eigentlichen Sitze des Scheins, wo manche Nebelbank und manches bald wegschmelzende Eis neue Länder lügt ...

It is the *land of truth* (an enchanting name), surrounded by a wide and stormy ocean, the very seat of illusion, where many a fog bank and many a swiftly melting iceberg feign new lands ...

Kritik der reinen Vernunft, A235 / B294 (1781 / 1787)

A land of truth in a sea of illusion



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Nebelbank

Weekend 1: honest uncertainty

wegschmelzendes Eis

Weekend 2: ground that is not there

[Land der Wahrheit](#)

Weekends 5 to 7: today

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Weekends 5 to 7: today

Weekend 1 was the fog. Weekend 2 was the ice. Now we go ashore.

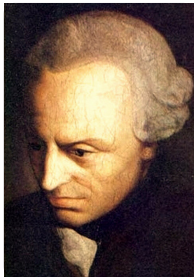
Four paradigms in science

	Ontological as it should be	Epistemic as it is
Thinking with brains	Probability 	Statistics 
Computing with computers	Computer science 	Machine learning 

What the five weekends behind us did

Weekend 0	NumPy, scikit-learn, PyTorch	the tools
Weekend 1	entropy, KL, variance, ensembles	dealing with variance
Weekend 2	confounding, selection, attribution	dealing with biases
Weekend 3	attacks, jailbreaks, privacy	defending models
Weekend 4	policy gradients, actor-critic, PPO	choosing the best next action

All the interest of my reason



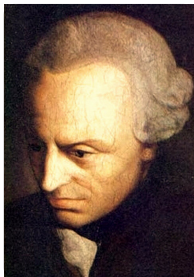
Immanuel Kant (c. 1790),
painter unknown

Alles Interesse meiner Vernunft (das spekulative sowohl, als das praktische)
vereinigt sich in folgenden drei Fragen:

- 1.
- 2.
- 3.

Kritik der reinen Vernunft, A805 / B833

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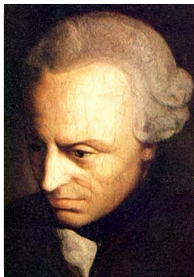
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What can I know?

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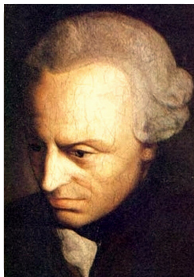
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What can I know? What ought I to do?

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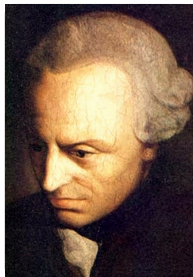
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Kant asked these of himself. We will ask them of the machine.

The same three questions, asked for the AI

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Was kann es wissen? → Weekend 5: RAG what it may know beyond its weights

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Was soll es tun? → Weekend 6: Agentic AI what it may do in the world

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The same three questions, asked for the AI

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Was soll es tun? → Weekend 6: Agentic AI what it may do in the world

Was darf es hoffen? → Weekend 7: Large Scale AI what it may become, for what we spend

darf, not *wird*: not what will happen, but what it is **entitled to expect**, once we have done our part.

wenn ich nun tue, was ich soll, was darf ich alsdann hoffen?